

Ethical Risk Assessment Brief

CS2 Match Outcome Prediction Using Machine Learning

1. Group Members

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2. Short Reminder of Research Topic

Which machine learning model should be employed to predict team wins in CS2 games with the highest possible accuracy? This forms the crux of our research problem. Statistics like win rate, average kills per round, economy, maps played, and head-to-head are the key factors being researched to determine the best possible model to predict the win rates with the highest possible accuracy. Player-level statistics like clutch rate, headshot percentage, and HLTV ratings are also considered as part of the overall investigation. Match data available on platforms like HLTV.org and other aggregators of CS2 statistics are the key sources considered for the overall investigation. Official sources like Valve and other esports data sources containing tournament records are also considered.

3. Identification of Ethical Risks

Risk 1: Privacy Violation and Player Re-identification

- **Who could be affected?**

Professional and semi-professional CS2 gamers whose in-game statistics, performance patterns, and behavioral data are publicly disclosed on websites such as HLTV.org

- **How could harm occur?**

Although the data is technically public, combining multiple features -such as round-level performance, map preferences, playtime patterns, and communication metadata -can enable re-identification of players who use pseudonyms or who have not explicitly consented to use of their data in predictive models. This could potentially reveal detailed behavioral profiles that gamers may not have intended to share in this aggregated manner.

- **When might this occur?**

This risk is most likely to occur if our model or dataset is published online or made public, or if someone uses the model to develop profiling tools for hiring, betting, or competitive intelligence without the consent of the players.

Risk 2: Reinforcement of Stereotypes and Discriminatory Labeling

- **Who might be affected?**

Players and teams from less represented countries (South America, CIS, Southeast Asia) that might be less represented due to the dominance of Western European and North American leagues.

- **How could harm occur?**

This means that if the training set favors the top-tier European/North American teams, the model will systematically assign lower probabilities of winning to teams from other regions - not because they are weaker teams, but because the training set does not fairly represent them. If the output of the model is used for scouting, seeding a tournament, or reporting on the media, it could exacerbate the inequality already seen in esports.

- **Under what circumstances might this happen?**

This is especially likely in a situation where data collection in a dataset is focused on prominent HLTV-ranked events, in which a small set of regions tends to dominate. Teams with fewer matches will have a more noisy feature representation, causing it to underestimate their potential.

Risk 3: Misuse of Predictions for Gambling or Match Fixing

- **Who might be harmed?**

Casual esports spectators, minors engaging in unregulated skin betting markets, and professional esports players, whose reputation may be tarnished based upon the association of divergent prediction results with suspicious betting patterns.

- **How might the harm occur?**

A prediction model published for public consumption may be leveraged by gambling organizations to establish odds that unfairly take advantage of less informed gamblers, or by match fixers to identify the most lucrative matches for manipulation. Professional esports players may unfairly be accused of match fixing if their win probability is consistently divergent from the prediction results.

- **Under what circumstances might this happen?**

This risk is realized if the model or its output is published without appropriate disclaimers, disseminated in communities where skin gambling is common, or incorporated into automated odds determination systems.

Risk 4: Misinterpretation of Model Confidence as Certainty

- **Who could be harmed?**

The potential users of this risk are esports analysts, coaches, tournament organizers, and team sponsors.

- **How could harm occur?**

The results of the machine learning model are based on probabilities, but users may misinterpret the results as certainty. For example, a coach may misinterpret an 85% probability of winning as a guarantee and end up making wrong decisions before the game. On the other hand, a sponsor may withdraw sponsorship from a team that has been performing poorly according to the model, without realizing that the results are based on data imbalance, not the actual performance of the team.

- **Under what circumstances might this happen?**

This is likely when presenting results without appropriate uncertainty quantification, or when model evaluation metrics are not clearly communicated to non-technical people.

4. Consent and Transparency Plan

In the context of the research project undertaken by the authors, the entire research relies on secondary data from publicly accessible platforms such as HLTV.org or the official CS2 statistics databases. There is no need to collect any data from the general public or any human subjects for the research undertaken by the authors. No personal messages or records need to be accessed for the research.

It is to be noted that the platforms mentioned above allow users to view aggregated match statistics for research purposes. Therefore, no informed consent from the subjects or the players is legally necessary for the research undertaken by the authors.

In order to maintain transparency, we shall:

- Document all the data sources we used appropriately by acknowledging them at the end of the report.
- Avoid the collection or storage of any personal identifiable information (PII) not absolutely necessary for the analysis.
- Ensure that we aggregate the results for the players at the team level to minimize the chances of any particular player being singled out.
- A disclaimer should be included with the publicly shared output. The disclaimer will be the following: "This model was developed exclusively for academic research purposes. The output of this model should not be used for any form of commercial forecasting, sports betting, player recruitment, or competitive intelligence. The authors accept no responsibility for any decisions made on the basis of these predictions." This disclaimer will be included on the title page of the report or at the beginning of the publicly shared code repository.
- Given that we are working with secondary public information, the professional players (participants) will not be notified about the research, and the participation will not be actively voluntary for the context of the research. Therefore, there will be no means for the players to withdraw their past public match statistics from the dataset.

5. Data Protection Plan

- **What data will be stored?**

Aggregated team and player stats such as win rate, kill death ratio, map score, etc. This data will be scraped/downloaded from public sources. No private user information, account credentials, or communications will be stored.

- **How will it be stored?**

Data will be stored locally on the personal academic device of the members of the group using password-protected local storage, and additionally in a restricted access

repository on a cloud storage platform approved by the university, such as OneDrive or Google Drive with restricted sharing capabilities.

- **Who will have access?**

Only the three group members and the instructor will have access to the information. There will be no sharing of the information with outsiders without a review.

- **How long will it be kept?**

The data will be kept for the length of the academic course. All the raw data files will be deleted within 30 days after the final grade submission.

- **How will unauthorized access be prevented?**

Files will be encrypted at rest with standard OS-level file encryption (BitLocker or FileVault). The repository will be set to “private” if used. No raw data files will be shared through public links or email as attachments or through unencrypted channels. Usernames for the player will be replaced with anonymous designators (Player_A, Player_B) before they are included in any shared analysis documents.

6. Bias and Fairness Considerations

- **Could the dataset underrepresent some groups?**

Yes. The HLTV dataset is biased toward the best European and North American events. The underrepresentation of the Oceanic, Middle Eastern, and some Southeast Asian teams is huge. This would make the model perform poorly in those areas. It would essentially learn that the geographic location is a factor for losing, which is a spurious correlation, not a reflection of actual team quality.

- **Would the model function differently with different groups?**

Yes. It is expected that the model will be more accurate with well-represented teams with more historical matches and less so with new or regional teams. This unevenness could cause the model to be not only less useful but also misleading with certain groups.

- **How will we verify that fairness is achieved?**

We can verify fairness by checking the accuracy of the model for different regions, like the EU, NA, CIS, and APAC. In addition to that, we can verify whether the error rates during predictions differ from region to region and report the same as part of the final analysis.

- **Could the results perpetuate inequality or reinforce existing stereotypes?**

There is a possibility that the results may perpetuate inequality or reinforce existing stereotypes. If the model is trained on data mainly from European competitions and concludes that the chances of winning are lower for non-European teams, it may be misinterpreted as an inherent inferiority factor rather than a biased data collection process. This factor will be taken into consideration while presenting the discussion part of the report.

7. Risk Mitigation Strategy

Identified Risk	Mitigation Action
Risk 1: Player re-identification	Replace player usernames with anonymized identifiers in all shared documents. Refrain from sharing the datasets. Aggregate player statistics at the team level before reporting.
Risk 2: Regional stereotyping and discriminatory labeling	Conduct stratified sampling for balanced representation. Assess the accuracy of the model for different regions. Provide explicit caveats for the report regarding the limitations of the data imbalance.
Risk 3: Misuse in gambling or match manipulation	Only submit the model for the academic submission. Provide a disclaimer for the model prohibiting the use for commercial purposes, such as gambling.
Risk 4: Misinterpretation of model confidence	Report the results probabilistically. Use calibration metrics to verify that the predictions made by the model correlate with actual outcomes. Calibration will be performed using the reliability diagram and Brier score. Avoid the use of language such as "the model predicts X will definitely win."

8. Ethical Reflection

If such an application of Counter Strike 2 match outcome prediction models like ours is ever implemented at a national or even global level, for example, in official match applications or live broadcast applications, a whole array of unforeseen consequences may arise. First and foremost, teams from less represented regions may be systematically neglected or even denied match invitations or sponsorships if their predicted win probability is consistently lower than others due to data imbalance issues.

Secondly, if such an application is implemented on a larger scale, the competitive scene of CS2 may undergo a significant change. For example, if teams and coaches only rely on statistics that our model clearly identifies as the best way to win, the direction of the entire competitive scene may change towards a more "safe" but possibly repetitive direction.

Lastly, if such an application is ever implemented in betting systems or applications, even unintentionally, it may inadvertently increase the advantage of gambling or market manipulation, which may adversely affect not only the competitive scene but also the financial situation of bettors. These concerns underscore that machine learning tools in competitive sports carry societal responsibilities that extend well beyond prediction accuracy.